

LIBRARY MANAGEMENT SYSTEM

Python Project Work

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Project Title	Library Management System
Programming Language	Python

1. Introduction

The Library Management System is a simple Python-based project developed to manage basic information about books in a library. The system allows the user to search for a book and immediately view its book number, location, and available stock. It also provides options to add new books and display all available books.

2. Objectives

- Store basic information about library books.
- Search for a book quickly using its name.
- Display the book number, shelf location, and available stock.
- Allow new books to be added to the system.
- Display all books stored in the library.

3. Features

- Search Book
- Add Book
- Show All Books
- Display Book Number
- Display Book Location
- Display Available Stock
- Exit Option

4. Technologies Used

Programming Language: Python

Concepts: Dictionary, loops, conditional statements, input/output and menu-driven programming.

5. Working of the Project

When the program starts, a menu is displayed. The user can select an option by entering a number. If the user searches for a book, the program checks the library dictionary. If the book exists, its name, number, location, and stock are displayed immediately. The user can also add a new book by entering its details.

6. Sample Output

===== LIBRARY MANAGEMENT SYSTEM =====

1. Search Book
2. Add Book
3. Show All Books
4. Exit

Enter your choice: 1

Enter book name: Python Programming

Book Found!

Book Name : Python Programming

Book Number : B001

Location : Shelf A1

Available Stock : 5

7. Python Source Code

```
# Library Management System

library = {
    "Python Programming": {
        "location": "Shelf A1",
        "number": "B001",
        "stock": 5
    },
    "Data Structures": {
        "location": "Shelf A2",
        "number": "B002",
        "stock": 3
    },
    "Computer Networks": {
        "location": "Shelf B1",
        "number": "B003",
        "stock": 4
    },
    "Engineering Mathematics": {
        "location": "Shelf B2",
        "number": "B004",
        "stock": 6
    }
}

while True:
    print("\n===== LIBRARY MANAGEMENT SYSTEM =====")
    print("1. Search Book")
    print("2. Add Book")
    print("3. Show All Books")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        name = input("Enter book name: ")

        if name in library:
            book = library[name]

            print("\nBook Found!")
            print("Book Name :", name)
            print("Book Number :", book["number"])
            print("Location :", book["location"])
            print("Available Stock :", book["stock"])
        else:
            print("Book not found!")

    elif choice == "2":
        name = input("Enter book name: ")
        number = input("Enter book number: ")
        location = input("Enter book location: ")
        stock = int(input("Enter stock: "))

        library[name] = {
            "number": number,
            "location": location,
            "stock": stock
        }

        print("Book added successfully!")

    elif choice == "3":
        print("\n===== ALL BOOKS =====")
```

```
        for name, book in library.items():
            print("\nBook Name :", name)
            print("Book Number :", book["number"])
            print("Location :", book["location"])
            print("Stock :", book["stock"])

    elif choice == "4":
        print("Thank you for using Library Management System!")
        break

    else:
        print("Invalid choice!")
```

8. Conclusion

The Library Management System successfully demonstrates how Python can be used to manage simple library records. The project is easy to use and provides quick access to book location, book number, and stock information. The system can be further expanded by adding student records, book issue and return features, due dates, fine calculation, and database storage.

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